

YOUSIF KHALID YOUSIF

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PROFESSIONAL SUMMARY

Software Engineer and Mobile App Developer specializing in building scalable, modular, and high-performance cross-platform applications using Flutter. Highly focused on translating complex business requirements and modern visual identities into highly responsive, dynamic frontend architectures. Adept at implementing clean architecture patterns and robust state management to support expanding "Super App" ecosystems. Equipped with strong secondary capabilities in backend infrastructure and edge AI integration, enabling a comprehensive understanding of end-to-end data pipelines while maintaining a primary focus on delivering seamless, component-driven user experiences.

CORE SKILLS

Mobile Frontend Architecture: Flutter, Dart, Clean Architecture, UI/UX Engineering, Widget-Driven Layouts

Languages: Arabic (Native), English (Excellent Level)

State Management & Data Flow: Bloc, AsyncValue, Client-Side Cache Systems

Secondary Technical Assets: Supabase (PostgreSQL), Edge AI (TFLite), Docker, Nginx, Python

KEY TECHNICAL PROJECTS

Super Manga — High-Throughput Content Distribution Platform *Flutter, Bloc, Supabase, Docker, Nginx, Python*

- Dynamic Frontend Orchestration:** Built a clean-architecture Flutter application utilizing Bloc for robust asynchronous state handling (AsyncValue), ensuring seamless UI performance across datasets containing 100,000+ graphic assets.
- High-Performance UX Execution:** Integrated a customized sliding-window pre-fetching system on the mobile UI to achieve zero-latency progressive rendering for high-resolution images, significantly optimizing image-heavy user journeys.
- Decoupled Architecture:** Designed a modular layout separating relational metadata from heavy binary streams, deploying a custom Dockerized Nginx engine on a Linux VPS to serve static assets and eliminate expensive cloud storage overhead.
- Automated Data Engineering:** Developed a convention-over-configuration pipeline in Python using the Linux directory structure as a data contract to automatically sanitize and structure flat asset folders into relational database tables.

Crop Care — AI-Driven Agricultural Diagnostic Mobile Application *Flutter, Dart, TensorFlow Lite (TFLite), MobileNetV2, Python*

- Scalable Interface Design:** Engineered a cross-platform Flutter application built on strict clean architecture principles, featuring deterministic camera and gallery image pipelines for seamless real-time data ingestion.
- Edge AI Integration:** Optimized and converted a dense Keras deep learning model into a lightweight TensorFlow Lite (.tflite) format, embedding a self-contained in-graph preprocessing layer to allow completely offline inference on resource-constrained mobile devices.
- Data Pipeline Optimization:** Mitigated severe data imbalance across minority plant classes during the R&D phase by implementing Categorical Focal Loss and inverse frequency class weighting, boosting model validation accuracy to 93% without compromising interface responsiveness.

EDUCATION

Degree in Computer Engineering *Focus on Software Architecture, Full-Stack Systems Integration, and Mobile Engineering*